

PAPER PRESENTATION · NEURIPS 2025

Transformers Provably Learn Chain-of-Thought Reasoning with Length Generalization

Trained on two-step chains, a one-layer transformer provably solves chains polynomially longer, when the group action allows it.

Yu Huang*, Zixin Wen*, Aarti Singh, Yuejie Chi, Yuxin Chen · UPenn, CMU, Yale · *equal contribution

Presented by Anurag · University of Waterloo

What is now proven about learned chain-of-thought

SIMPLY TRANSITIVE ACTIONS

$$2 \longrightarrow O(d^{c^*})$$

Trained on chains of length at most 2, accurate on every chain up to length $O(d^{c^*})$.

Theorem 4.1

SYMMETRIC ACTIONS

$$2^k \longrightarrow 2^{k+1}$$

Trained up to length 2^k , accurate at length 2^{k+1} : one guaranteed doubling.

Theorem 5.1

RECURSIVE SELF-TRAINING

$$\Theta(\log d) \text{ rounds}$$

Self-labeled doubling reaches every length $L \leq |X|$, the maximum the setting allows.

Corollary 5.1

The first optimization guarantee that a constant-depth (here: one-layer) transformer learns an NC^1 -complete problem with CoT, which is beyond TC^0 unless $TC^0 = NC^1$.

Why practitioners should care

Seen in practice	What the paper explains	Practical reading (ours)
Reasoning models must run longer chains than any training trace	Extrapolation is limited by retrieval robustness, not by the per-step computation	Track attention mass on the clauses a step needs, as a function of context length
"Context rot": accuracy decays as the context grows	Irrelevant clauses dilute softmax attention; leaked mass grows linearly in length	Tasks full of look-alike distractors need sharper attention than short-context training induces
Training on a model's own longer outputs extends its reach	Self-labeled doubling provably extends the horizon; $\Theta(\log d)$ rounds reach the maximum	Double the length per round and keep self-labels near-perfect, so errors do not compound

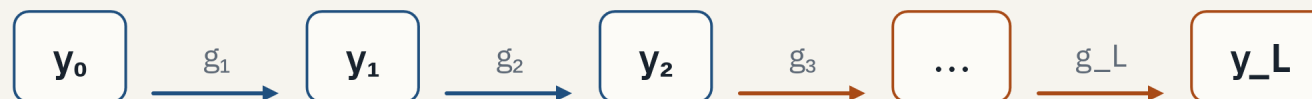
The right-hand column is interpretation. Every theorem concerns a one-layer transformer on synthetic LEGO state-tracking tasks.

The problem: does learned CoT extrapolate in length?

State tracking. A group G acts on a set of states Y . Given a start state y_0 and actions $g_1, \dots, g_L$, output the final state

$$y_L = g_L \circ g_{L-1} \circ \dots \circ g_1 (y_0)$$

Chain-of-thought emits $y_1, y_2, \dots, y_L$ one step at a time; every intermediate state is supervised during training.



Train: 2 steps (or up to 2^k with self-training)

Test: far longer chains, up to $O(d^{c*})$ or $|X|$

Question. When does gradient descent on short chains yield per-step accuracy ≈ 1 on much longer chains, and why does it fail when it fails?

State tracking is the serial core of reasoning

Known: CoT makes it expressible

Without CoT, constant-depth log-precision transformers compute only TC^0 functions (Merrill & Sabharwal 2023).

Tracking a non-solvable group such as S_5 is NC^1 -complete (Barrington 1986): out of reach unless $TC^0 = NC^1$.

With L CoT steps, a constant-depth transformer can express it (Li et al. 2024; Merrill & Sabharwal 2024).

Open: is it learned, and does it extrapolate?

Expressibility says a solution exists, not that gradient descent finds it.

Reasoning scales by running longer chains; CoT that does not extrapolate needs fresh supervision at every length.

State tracking sits under program execution, entity tracking, games and automaton simulation.

Why it is hard: three obstacles

01

Retrieval under dilution

Each step must find one relevant clause among $O(L)$. Softmax leaks mass to every irrelevant clause, so a model fit at $L = 2$ can fail at $L = 100$.

02

Coupled, non-convex dynamics

Softmax attention feeds a nonlinear FFN. Which neurons learn which group entries depends on the random initialization.

03

Out of distribution by design

Test lengths are never seen, so in-distribution generalization bounds are silent, and positional encodings can break extrapolation.

Without CoT, transformers track state through shortcuts whose depth grows with $\log L$ (Liu et al. 2023). CoT removes the depth problem and moves the burden onto retrieval.

Why it was open: each line of work covers one piece

Line of work	What it guarantees	What it leaves open
Expressivity of CoT · Feng et al. 2023; Merrill & Sabharwal 2024; Li et al. 2024	Constant-depth transformers with CoT can represent NC^1 -complete computations	Whether gradient descent finds them
Optimization with CoT · Kim & Suzuki 2025; Wen et al. 2025; Huang, Liang & Yang 2025; Huang, Wang & Lee 2025	Gradient descent learns parity, regular-language checks, multi-step linear regression	TC^0 tasks only; no lengths beyond training
Length-generalization theory · Huang et al. 2025; Golowich et al. 2025; Joshi et al. 2025	Extrapolation under idealized inference; sample complexity independent of chain length	No gradient-descent dynamics
Self-improvement · Lee et al. 2025	Large empirical length gains from training on own outputs	No theory

The approach: separate retrieval from computation

- 1 • Task.** LEGO state tracking, written as 5-token clauses (Zhang et al. 2022)
- 2 • Model.** One layer, no positional encoding: softmax attention, then a smoothed-ReLU FFN
- 3 • Training.** Gradient descent in stages: FFN first (computation), attention second (retrieval)
- 4 • Currency.** Attention concentration on the two clauses a step needs
- 5 • Dichotomy.** Free (simply transitive) versus non-free (symmetric) group actions
- 6 • Remedy.** Recursive self-training that doubles the length each round

Limitations, stated up front

One layer, one attention matrix

Fixed orthonormal embeddings; dimension at least the number of variables

Only 2 of the 25 attention blocks are trainable

Stage-wise training with frozen parts

Gradients of the population loss: no sample complexity

Synthetic task with tiny groups ($|G| \leq \text{polylog } d$)

One LEGO sentence, one reasoning step

PREDICATES, IN ANY ORDER

$$b = g_1(a), c = g_2(b), d = g_3(c)$$

ANSWERS, GENERATED AS COT

$$a = 2 \rightarrow b = 0 \rightarrow c = 3 \rightarrow d = 2$$

Group C_6 acting on $\mathbb{Z}_6$ with $g_1 = +4, g_2 = +3, g_3 = +5$

Clause	1	2	3	4	5
$b = g_1(a)$	b	g_1	a	.	.
$c = g_2(b)$	c	g_2	b	.	.
$a = 2$	.	.	.	a	2
$b = 0$ (next)	.	.	.	b	0

One step from the query "a = 2" (· marks a blank slot)

Attention matches a against slot 3 of predicates (finds $b = g_1(a)$) and slot 4 of answers (finds $a = 2$).

FFN copies b into slot 4 and writes $g_1(2) = 2 + 4 = 0 \pmod{6}$ into slot 5.

Contributions, and where each appears in this talk

Contribution	Statement	Slides
Theorem 4.1	Simply transitive actions: training at $L \leq 2$ gives per-step accuracy $\geq 1 - 1/\text{poly}(d)$ for every $L \leq O(d^{c^*})$	17, 25–26
Theorem 5.1	Symmetric actions: training up to 2^k gives accuracy at $2^{(k+1)}$; attention concentration $\geq 1 - \tilde{c}$ with $\tilde{c} < 0.01$	18, 27
Corollary 5.1	Recursive self-training reaches every $L \leq X $ after $\Theta(\log d)$ rounds	18, 28
Mechanism	Attention concentration ties retrieval robustness to the algebra of the action	22–28
Complexity reading	Gradient descent with CoT learns an NC^1 -complete task with a one-layer transformer	21
Experiments	C_6 versus S_5 length curves, self-training curves, attention maps at convergence	32

The data: LEGO sentences generated by a group action

DEFINITION 2.2 · LEGO LANGUAGE (ZHANG ET AL. 2022; PRELIMINARY, NOT NEW)

Variables X , actions G (each a map $Y \rightarrow Y$), values Y . Predicate clause $x = g(x')$; answer clause $x = y$. A length- L sentence lists the chain, then its answers:

$$x_n = g_n(x_{n-1}), n \in [L]; \quad x_0 = y_0, \quad x_n = y_n, n \in [L']$$

ASSUMPTION 3.2 · SAMPLING

$x_0, \dots, x_L$ uniform from X without replacement

$y_0 \sim \text{Unif}(Y)$; $g_1, \dots, g_L \sim \text{Unif}(G)$, independent

$y_i = g_i(y_{i-1})$, so every answer is consistent

The truncation $D^{(L,L')}$ keeps answers $0, \dots, L'$

ASSUMPTION 3.1 · ASYMPTOTIC REGIME

$$d \rightarrow \infty, \quad |\mathcal{X}| \rightarrow \infty, \quad |\mathcal{G}| \leq \log^{C_0} d, \quad C_0 \in [1, 100)$$

Groups are tiny and variable names plentiful. Distinct variables cap the chain: $L + 1 \leq |X| = \Theta(d)$, where d is the vocabulary size.

Two kinds of action: the axis of the whole paper

SIMPLY TRANSITIVE · FREE AND TRANSITIVE

$$\forall y, y' \in \mathcal{Y} \quad \exists! g \in \mathcal{G} : g \cdot y = y'$$

Exactly one action takes any state to any other, so $|G| = |Y| = n_y$.

Assumption 4.1: $n_y \in [\Omega(\log \log d), \log d]$.

Examples: C_n on $\mathbb{Z}_n$, and S_n acting on itself, which is the S_n word problem.

SYMMETRIC · TRANSITIVE, NOT FREE

$$\mathcal{G} = \text{Sym}(\mathcal{Y}) : |\{g : g \cdot y = y'\}| = (n_y - 1)!$$

Many actions realize each transition; stabilizers are non-trivial.

Assumption 5.1: $n_y = \Theta(\log \log d / \log \log \log d)$, so $|G| = n_y! = \text{polylog } d$.

Example: S_n permuting $\mathbb{Z}_n$, NC¹-complete once $n_y \geq 5$ (Barrington).

The same group can sit on both sides: S_n acting on itself is simply transitive, S_n permuting $\mathbb{Z}_n$ is not. The closing experiment uses exactly this pair.

The model: one attention layer, one FFN, no positions

EMBEDDING

$v \mapsto e_v \in \mathbb{R}^d$ orthonormal, $(v_1, \dots, v_5) \mapsto \mathbf{Z} = (e_{v_1}, \dots, e_{v_5}) \in \mathbb{R}^{5d}$

ATTENTION

$$\text{Attn}_{j \rightarrow k} = \frac{\exp(\mathbf{Z}_j^\top \mathbf{Q} \mathbf{Z}_k)}{\sum_r \exp(\mathbf{Z}_j^\top \mathbf{Q} \mathbf{Z}_r)}, \quad \text{Attention}(\mathbf{Q}, \mathbf{Z}) = \sum_k \text{Attn}_{\text{ans}, L' \rightarrow k} \mathbf{Z}_k$$

Query = last answer clause; causal mask; Q stands for $\mathbf{W}_Q^\top \mathbf{W}_K$; no value or output matrix.

FFN

$$\text{FFN}_{i,j}(\mathbf{X}) = \sum_{r \in [m]} \text{sReLU}(\langle \mathbf{W}_{i,j,r}, \mathbf{X} \rangle + b_{i,j,r}), \quad i \in [5], j \in [d]$$

sReLU behaves like $x^q / (q \cdot q^{(q-1)})$ near 0 and is linear beyond q ; q is a large even constant, $q = \Theta(1/\text{polylog } d)$; biases are fixed.

OUTPUT

Five parallel softmaxes over the vocabulary, one per slot of the next clause; logits clipped at $C_B \log d$ (Assumption 3.4).

TRAINABLE

$\mathbf{W}$, and only blocks (4,3) and (4,4) of $\mathbf{Q}$: the query's variable slot against the keys' slots 3 and 4 (Assumption 3.5).

The training rule: staged GD on the population loss

NEXT-CLAUSE LOSS WITH TEACHER FORCING; LOSS_I SCORES SLOT I ONLY

$$\text{Loss}^{L,L'}(F) = \mathbb{E}_{Z \sim \mathcal{D}^{L,L'}} \left[-\log p_F(Z_{\text{ans},L'} \mid Z^{L,L'-1}) \right], \quad \text{Loss}^L = \sum_{L'=1}^L \text{Loss}^{L,L'}$$

INITIALIZATION (ASSUMPTION 3.3)

$$\mathbf{Q}^{(0)} = \mathbf{0}, \quad \mathbf{W}_{i,j,r}^{(0)} \sim \mathcal{N}(0, \sigma_0^2 \mathbf{I}), \quad \sigma_0 = d^{-1/2}, \quad b_{i,j,r} = \sigma_0 \log d \text{ (fixed)}$$

Stage 1 · FFN

$$\mathbf{W} \leftarrow \mathbf{W} - \eta \nabla_{\mathbf{W}} \text{Loss}^1, \quad T_1 = \tilde{O}(d/(\eta \sigma_0^{q-2})) \quad \mathbf{Q} = \mathbf{0}, \text{ so attention is uniform}$$

Stage 2 · attention

$$\mathbf{Q} \leftarrow \mathbf{Q} - \eta \nabla_{\mathbf{Q}} \sum_{\ell=1}^2 \text{Loss}_5^{2,\ell}, \quad T_2 = \tilde{O}(\text{poly}(d)/(\eta \sigma_0)) \quad \mathbf{W} \text{ frozen; Algorithm 2 scores only the second answer}$$

Round $k \geq 2$ · Alg. 2

$$\mathbf{Q} \leftarrow \mathbf{Q} - \eta \nabla_{\mathbf{Q}} \text{Loss}_{\tilde{F}(k),5}^{2^k,2} \quad \text{while} \quad \text{Loss}_{\tilde{F}(k),5}^{2^k,2} > d^{-E_1} \quad \mathbf{W} \text{ frozen; labels by greedy decoding of the previous model}$$

$\eta = 1/\text{poly}(d)$. Algorithm 1: stages 1–2. Algorithm 2: stages 1–2, then self-training rounds.

The evaluation: teacher-forced, per-step exact match

$$\text{Acc}_L(F) = \frac{1}{L} \sum_{0 \leq L' < L} \mathbb{E}_{Z^L \sim \mathcal{D}^L} \mathbb{E}_{\hat{Z} \sim p_F(\cdot | Z^{L,L'})} \left[\mathbf{1} \{ \hat{Z}_{\text{ans}, L'+1} = Z_{\text{ans}, L'+1} \} \right]$$

PER STEP

The prefix is ground truth, and all five tokens of the next answer clause must match.

WHOLE CHAIN (OURS)

$$\Pr[\text{all } L \text{ free-running steps correct}] \geq 1 - L(1 - \text{Acc}_L)$$

Correct steps reproduce the true prefix, so a union bound applies: $1 - 1/\text{poly}(d)$ per step gives at least $1 - L/\text{poly}(d)$ per chain, meaningful because $L \leq |X| = \Theta(d)$.

MECHANISM METRIC

$$\text{Attn}_{\text{ans}, \ell \rightarrow \text{pred}, \ell+1} + \text{Attn}_{\text{ans}, \ell \rightarrow \text{ans}, \ell}$$

Attention concentration: the query's mass on the two clauses a step needs. Both theorems bound it from below.

What kind of result is this?

Category	Verdict	Reason
Representation	Background	Expressibility of state tracking with CoT is already known
Computational complexity	Framing	NC ¹ -completeness interprets the result; no new circuit bound
Trainability / optimization	Core	Explicit GD trajectory: feature learning in W, then attention in Q
Learnability	Yes, by GD	Population-loss GD reaches $1 - 1/\text{poly}(d)$; no sample-size statement
Generalization	Yes, in length	Accuracy at lengths never trained on (OOD), not i.i.d. generalization
Certification	No	No verifier, certificate or robustness guarantee

Verdict: an optimization theorem whose payoff is out-of-distribution length generalization.

Read "learn" as "gradient descent on the population loss reaches it", not PAC learnability; read "generalize" as "extrapolate in L".

Theorem 4.1: simply transitive actions

HYPOTHESES

Assumptions 3.1 to 3.5 and 4.1; some constant $0 < c^* < 1$; $n_y < m \ll \log^2 d$

$F = F^{(T_1+T_2)}$ from Algorithm 1, $\eta = 1/\text{poly}(d)$, $T_1 = \tilde{O}(d/(\eta\sigma_0^{q-2}))$, $T_2 = \tilde{O}(\text{poly}(d)/(\eta\sigma_0))$

(1) LENGTH GENERALIZATION

$$\text{Acc}_L(F) \geq 1 - \frac{1}{\text{poly}(d)} \quad \text{for every } L \leq O(d^{c^*})$$

(2) ATTENTION CONCENTRATION

$$\text{Attn}_{\text{ans}, \ell \rightarrow \text{pred}, \ell+1} + \text{Attn}_{\text{ans}, \ell \rightarrow \text{ans}, \ell} \geq 1 - O(d^{-c^*}), \quad \ell \in \{0, 1\}$$

Trained only at $L \in \{1, 2\}$

c^* exists; its value is not given

Uniform over every $L \leq O(d^{c^*})$

Theorem 5.1 and Corollary 5.1: symmetric actions

HYPOTHESES

Assumptions 3.1, 3.2, 5.1 (data) and 3.3 to 3.5 (network); $n_y < m \ll \log^2 d$

for any $1 \leq k < \log_2 |\mathcal{X}|$: $F^{(T_k)}$ trained by Algorithm 2 up to $L_k = 2^k$, $T_k = O(\text{poly}(d)/\eta)$

(1) ONE DOUBLING $\text{Acc}_{2^{k+1}}(F^{(T_k)}) \geq 1 - \frac{1}{\text{poly}(d)}$

(2) CONCENTRATION $\text{Attn}_{\text{ans}, \ell \rightarrow \text{pred}, \ell+1} + \text{Attn}_{\text{ans}, \ell \rightarrow \text{ans}, \ell} \geq 1 - \tilde{c}, \quad \tilde{c} < 0.01, \quad \ell \in \{0, \dots, L_k - 1\}$

COROLLARY 5.1 $K = \Theta(\log d) : \quad \text{Acc}_L(F^{(T_K)}) \geq 1 - \frac{1}{\text{poly}(d)} \quad \text{for every } L \leq |\mathcal{X}|$

Only constant concentration, $1 - \tilde{c}$: one doubling per round, never d^{c^*} in one shot.

What each guarantee depends on

Quantity	How it enters the theorems
Vocabulary size d	The asymptotic parameter: every bound holds as $d \rightarrow \infty$; the embedding dimension is $5d$
Sequence length L	$\leq O(d^{c^*})$ in Thm 4.1; $\leq 2^{(k+1)}$ after round k in Thm 5.1; $\leq X = \Theta(d)$ in Cor 5.1
Width m	$n_y < m \ll \log^2 d$ smoothed-ReLU neurons per (slot, output token)
Depth and heads	One layer and one attention matrix Q ; nothing is claimed about deeper models
Precision	Exact real arithmetic; output logits clipped at $C_B \log d$, a log-precision scale
Conditioning	Orthonormal token embeddings; initialization $\sigma_0 = d^{-1/2}$; fixed bias $\sigma_0 \log d$
Task complexity	$ G \leq \log^{(C_0)} d$; $n_y \in [\Omega(\log \log d), \log d]$ or $n_y = \Theta(\log \log d / \log \log \log d)$
Sample size	None: every step uses the gradient of the population loss
Compute	$\eta = 1/\text{poly}(d)$; $\text{poly}(d)/\eta$ steps per stage; $\Theta(\log d)$ self-training rounds

Reading the statements precisely

c^* is existential

The horizon is $O(d^{c^*})$ for some unspecified $c^* \in (0,1)$; no rate in the training length.

Accuracy is per step

$1 - 1/\text{poly}(d)$ holds per step with the true prefix; a whole chain gets $1 - L/\text{poly}(d)$.

Lengths are in units of d

$d^{c^*} \ll |X| = \Theta(d)$: horizons are measured against vocabulary size, not training length.

Constant depth = one layer

The abstract's "constant-depth transformer" is a single attention + FFN block.

Tiny supervision

Thm 4.1 never trains past $L = 2$; a self-training round scores one value token: the second answer.

Population, not samples

Every gradient is of an expected loss; finite-sample effects are outside the theorems.

The NC^1 claim: what is proven and what it means

Proven

Gradient descent with CoT learns tracking for S_n on $\mathbb{Z}_n$ (Corollary 5.1).

As stated, Assumption 4.1 also admits S_n acting on itself, the word problem, so Theorem 4.1 covers an NC^1 -complete case too.

Both are NC^1 -complete once $n \geq 5$ (Barrington 1986).

What it means

Without CoT, log-precision constant-depth transformers are in TC^0 , so they cannot solve these unless $TC^0 = NC^1$.

With L CoT steps a solution exists (Li et al. 2024).

New here: gradient descent finds one.

Caveat

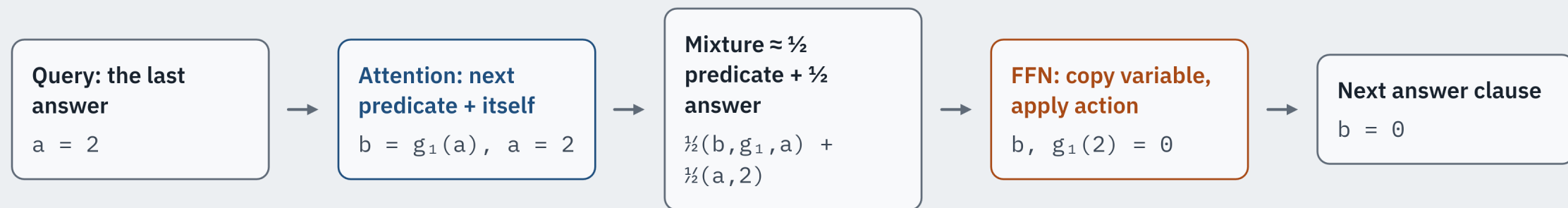
Each CoT step is one retrieval plus a lookup in a table of $|G| \cdot |Y| = \text{polylog } d$ entries.

The serial depth comes from the chain, not the network.

What gradient descent must learn is robust retrieval, not an NC^1 circuit.

The class describes the no-CoT baseline; the new content is the optimization guarantee.

One reasoning step = retrieve, then compute



A · Copy map (§6.1.1)

The FFN learns to copy the next variable into slot 4.

B · Action table (§6.1.2)

The FFN learns $g(y)$ for every action–state pair.

C · Concentration (§6.2)

Attention learns to match the current variable.

Key trick. With $Q = 0$ and a two-clause context, attention is exactly $\frac{1}{2}$ and $\frac{1}{2}$, so the FFN trains on a clean superposition; it is then frozen while Q learns.

The FFN learns to copy the next variable

PRE-ACTIVATION OF A SLOT-4 NEURON ON THE ONE-STEP TASK

$$\Lambda_{4,j,r}(Z^{1,0}) = \frac{1}{2} \langle \mathbf{W}_{4,j,r}, \mathbf{Z}_{\text{pred},1} \rangle + \frac{1}{2} \langle \mathbf{W}_{4,j,r}, \mathbf{Z}_{\text{ans},0} \rangle + b_{4,j,r}$$

The next variable x_1 appears twice: in slot 1 of the input predicate, and as the slot-4 target.

So the diagonal weights (slot 1, variable s , output s) get strictly larger gradient updates than any other entry.

Weights toward wrong variables, values and actions stay $o(1)$.

End state: a copy map from slot 1 to slot 4, for every variable.

$$\langle \mathbf{W}_{4,s,r,1}, e_s \rangle \text{ grows; } \quad \langle \mathbf{W}_{4,s,r,p}, e_{s'} \rangle = o(1) \text{ for } (p, s') \neq (1, s)$$

Why it extrapolates

The copy map acts variable by variable and never sees positions or other clauses, so nothing in it depends on L .

Running example: from $\frac{1}{2}(b, g_1, a) + \frac{1}{2}(a, 2)$, slot 4 outputs b .

The FFN learns the action table by competition

COMBINATIONS FOR OUTPUT VALUE j , AND THE FEATURE A NEURON ASSIGNS TO ONE

$$\Phi_j^* = \{(g', y') \in \mathcal{G} \times \mathcal{Y} : \tau(g'(y')) = j\}, \quad V_{j,r}(\phi) = \frac{1}{2} (\langle \mathbf{W}_{5,j,r,2}, e_g \rangle + \langle \mathbf{W}_{5,j,r,5}, e_y \rangle)$$

Phase I · emergence

$$V_{\psi}^{(t+1)} \geq V_{\psi}^{(t)} + \tilde{\Omega}(\eta) (b + V_{\psi}^{(t)} \pm o(\mu))^{q-1}$$

In sReLU's polynomial regime, growth is a tensor-power iteration: the feature with the largest initial value wins, giving an implicit curriculum.

Phase II · growth and cancellation

The winner keeps growing. Confounders sharing one component with it, (g', y) or (g, y') , are cancelled through a stationarity property of the gradient.

Outcome (simply transitive). For each value j and each pair (g, y) with $g(y) = j$, exactly one neuron specializes; n_y^2 neurons activate in all, the rest stay near initialization.

Attention learns to match the current variable

$$\mathbf{z}_{\text{ans},\ell}^\top \mathbf{Q} \mathbf{z}_k = e_{x_\ell}^\top \mathbf{Q}_{(4,3)} e_{\text{slot}_3(k)} + e_{x_\ell}^\top \mathbf{Q}_{(4,4)} e_{\text{slot}_4(k)}$$

The current variable sits in slot 3 of the next predicate and slot 4 of the current answer: one signal for both trainable blocks.

With the FFN frozen, gradient descent raises the matching scores while leaked attention still costs loss.

Result (Theorem 4.1): concentration $\geq 1 - O(d^{-c^*})$ after training at $L = 2$ only.

Running example

Query $a = 2$: the score is high exactly for $b = g_1(a)$, whose slot 3 is a , and for $a = 2$ itself, whose slot 4 is a .

Intuition (ours): with orthonormal codes, every non-matching clause sits a learned gap Δ below the matching ones.

Why concentration buys length: dilution

$$\varepsilon(L) = \frac{N(L) e^{-\Delta}}{2 + N(L) e^{-\Delta}} \approx L e^{-\Delta}, \quad N(L) \approx 2L$$

$$e^{-\Delta} = O(d^{-c^*}) \implies \varepsilon(L) = O(L d^{-c^*}), \text{ small iff } L \ll d^{c^*}$$

Length L	2	$d^{c^*} / 100$	$d^{c^*} / 10$	d^{c^*}
Leaked mass $\varepsilon(L)$	$\approx 2 \cdot d^{(-c^*)}$	≈ 0.01	≈ 0.09	0.5

The FFN is position-free: it reads only the attention-weighted mixture. Accuracy survives while $\varepsilon(L)$ stays below the FFN's tolerance; Theorem 4.1 makes this rigorous.

Back-of-envelope (ours): equal scores among relevant and among irrelevant clauses; $N(L) = (L - 1)$ predicates + ℓ earlier answers.

Harmless distractors stall attention at a constant

Distractor: an irrelevant clause whose action also sends the current state to the right next state.

For S_n on $\mathbb{Z}_n$, $(n_y - 1)!$ actions realize each transition; a simply transitive action has exactly one.

Attending to a distractor still yields the correct value, so the loss gets small while attention is still diffuse.

Training stops at concentration $\geq 1 - \tilde{c}$, $\tilde{c} < 0.01$: enough for $2 \times L_{\text{train}}$, not for d^c .

Empirically, relevant attention for S_5 dilutes as test length grows, while C_6 's holds (full version, §7).

Our note: a question to raise

A uniformly random irrelevant clause gives the right transition with probability $1/n_y$ in both families (C_n : $g' = g$; S_n on $\mathbb{Z}_n$: $g'(y) = g(y)$).

Frequency alone cannot separate them. The separation must come from how the learned FFN and the loss treat look-alike actions, which is where §6.2.2 should be checked.

Recursive self-training: each round buys one doubling



Why self-labels are safe

The previous round is already accurate at 2^k (Theorem 5.1), so its greedy labels are correct with probability $1 - 1/\text{poly}(d)$. No filtering is needed.

What is retrained

Only Q ; W stays frozen. Self-training sharpens retrieval; the action table from stage 1 never changes.

**GAP INTUITION
(OURS)**

$$\Delta_k \approx \log(2^k / \tilde{c}) = \Delta_{k-1} + \log 2, \quad K = \Theta(\log d) \Rightarrow \Delta_K \approx \log |\mathcal{X}|$$

The carrying assumption: abundant orthonormal codes

Every variable has its own orthonormal embedding, in dimension at least $|X| = \Theta(d)$. Retrieval is exact matching, so the only error is softmax leakage, set by one learned gap on a $\Theta(\log d)$ logit scale.

$$L_{\max} \approx e^{\Delta}, \quad \Delta \approx c^* \log d \implies L_{\max} \approx d^{c^*}$$

Block-sparse Q

The model is told which slots to compare (4 against 3, 4 against 4) and learns only how sharply.

Staged training

The FFN is learned under exactly uniform attention, then frozen while Q learns.

The authors' own flag

Footnote 4: the theory does not cover embedding dimension logarithmic in the problem length.

The formal model next to a modern LLM

Component	This paper	Modern LLM
Depth, heads	One layer, one attention matrix	Dozens of layers, many heads
Positions	None (causal mask only)	Rotary or other positional schemes
Tokens	One clause per position, five output softmaxes	One subword per position
Embeddings	Fixed, orthonormal, dimension $5d \geq 5 X $	Learned; width far below context length
Attention	Merged Q, no value map, 2 of 25 blocks trainable	Full query, key, value, output maps
FFN, stream	sReLU, fixed biases, no residual or norm	SwiGLU-type FFN, residuals, normalization
Optimization	Staged GD, population loss, $\eta = 1/\text{poly}(d)$	Joint AdamW on minibatches
Supervision	Teacher-forced step-level CoT, synthetic LEGO	Pretraining, SFT, RL on text and code

The correspondence is at the level of mechanism, not architecture.

What plausibly transfers, and what does not

Mechanism level: likely transfers

Retrieve-then-compute for each CoT step, as in the identical-token "association" heads seen in LEGO-trained transformers

Length failure as attention dilution: a measurable quantity

Doubling curricula on self-generated data

Quantitative level: does not transfer

The horizon d^c , which relies on one orthonormal code per variable

Staged training with frozen parts

One-layer computation of each step and teacher-forced step supervision

Carry the mechanism to real models; leave the numbers behind.

The experiments: what they measure

FIGURE 1A

Length generalization

C_6 against S_5 ; accuracy over lengths 5 to 160, training length dashed. C_6 stays accurate far past training; S_5 does not.

FIGURE 1B

Self-improvement on S_5

Ground truth at $L = 5$, then self-labeled rounds at $L = 10, 20, 40$. Each round shifts the accuracy curve to the right.

FIGURE 2

Attention at convergence

Heatmaps at $L = 5$ for C_6 and S_5 : two diagonal bands, from each answer to itself and to the next predicate.

Widom's test, passed: they measure sensitivity to the key parameter (length) and the predicted mechanism (attention), not only accuracy.

Not shown in Figures 1–2: lengths beyond 160, or how the horizon scales with d .

Ledger: theorem, evidence, conjecture, interpretation

Theorem, under all assumptions

Thm 4.1: horizon $O(d^{c^*})$ and concentration $1 - O(d^{-c^*})$, simply transitive actions

Thm 5.1: one doubling per round, concentration $1 - \tilde{c}$, S_n on $\mathbb{Z}_n$

Cor 5.1: every $L \leq |X|$ after $\Theta(\log d)$ rounds

Experimental only

That symmetric training fails beyond a constant factor: Thm 5.1 and Cor 5.1 state only the positive direction

Finite-d behavior: C_6 accurate to $L = 160$; S_5 attention diluting with length

Conjecture

Footnote 4: embedding dimension logarithmic in length might suffice with over-complete codes

Interpretation

"Algebraic structure governs length generalization", generalized from two action families

Links to context rot and to LLM self-improvement; "beyond TC⁰" as significance

Where the paper sits: the one empty cell

What is guaranteed	Fixed-length tasks	Serial state tracking	Length generalization
Expressivity		Liu et al. 2023; Merrill & Sabharwal 2024; Li et al. 2024	
Idealized or statistical learning	Joshi et al. 2025		Huang et al. 2025; Golowich et al. 2025
Gradient-descent optimization	Kim & Suzuki; Wen et al.; Huang, Liang & Yang; Huang, Wang & Lee (all 2025)	This paper	This paper
Empirical		Zhang et al. 2022 (LEGO); Li, Guo & Andreas 2025	Zhou et al. 2024; Lee et al. 2025

From TC^0 at fixed length to NC^1 with extrapolation

Paper	What it shows	This paper, relative to it
Kim & Suzuki ICLR 2025	One-layer model learns parity in one gradient step under teacher-forced CoT	Same step supervision; a non-solvable group, and guarantees at untrained lengths
Wen et al. ICLR 2025	CoT makes parity learnable from polynomially many samples via sparse attention	Makes sparse attention quantitative (concentration) and ties it to the horizon
Huang, Liang & Yang ICML 2025	Attention + linear layer on even pairs and parity check; max-margin bias	Nonlinear FFN that must learn an action table; staged, not joint, training
Huang, Wang & Lee ICLR 2025	CoT lets linear transformers run multi-step GD for in-context regression	A discrete serial task whose difficulty is retrieval, not regression

Assumptions changed: no positional encoding, clause-level tokens, block-sparse Q, frozen stages.

From existence and idealized learners to GD

Paper	What it shows	This paper, relative to it
Li et al.; Merrill & Sabharwal ICLR 2024	T CoT steps express size-T circuits; linear steps capture regular languages	From existence to learnability, for one NC ¹ -complete family, under teacher forcing
Huang et al. ICLR 2025	Length generalization for an idealized norm-regularized learner, absolute positions	Actual GD dynamics instead of an idealized learner, and no positions at all
Golowich et al. ICML 2025	Dependence on k earlier tokens enables extrapolation; motivates position coupling	Dependence is 2-sparse but content-addressed, so no positional device is needed
Joshi et al. COLT 2025	Time-invariant CoT: sample complexity independent of chain length	Time invariance is the position-free FFN; length enters only through dilution

Narrower scope than all four: one task family, one layer, and supervision at every step.

What the theory explains, and what it assumes away

Paper	What it shows	This paper, relative to it
Lee et al. ICML 2025	Filtered self-generated data; 10- to 100-digit addition; exponential gains over rounds	Proves one doubling per round, so exponential growth in rounds; no filter needed
Zhang et al. 2022 (LEGO)	Introduced the task; found heads that attend to identical tokens	Proves that matching pattern emerges under GD and quantifies it
Liu et al. ICLR 2023; Li, Guo & Andreas 2025	Without CoT: shortcuts or associative scans, with depth growing in length	CoT trades depth for retrieval; the failure mode becomes dilution

Changed assumption. Lee et al. filter because real self-labels are noisy; the theory instead assumes the previous round is already $1 - 1/\text{poly}(d)$ accurate at the doubled length.

In this model, the length horizon is set by retrieval, not computation

$$L_{\text{horizon}} \approx L_{\text{train}} \cdot \frac{\tau}{\epsilon_{\text{train}}}$$

Terms

ϵ_{train} : attention leaked at training length.
 τ : contamination the position-free FFN tolerates.

Simply transitive

$\epsilon_{\text{train}} = O(d^{-c^*})$ from $L_{\text{train}} = 2$, so the horizon is polynomial in d .

Symmetric

$\epsilon_{\text{train}} = \tilde{c}$, so a constant factor; self-training re-earns a small ϵ at every doubling.

The formula is our one-line synthesis of Theorems 4.1 and 5.1, not a statement from the paper.

The horizon is bought with embedding dimension

$$\underbrace{5d \geq 5|\mathcal{X}| > 5L}_{\text{embedding dimension}} \quad \text{and} \quad L \leq O(d^{c^*}) \implies d \geq \Omega(L^{1/c^*}) \gg L$$

Every guarantee assumes one orthonormal code per variable, in dimension $5d \geq 5|\mathcal{X}| > 5L$. Theorem 4.1's horizon $L \leq O(d^{c^*})$ then forces $d \geq \Omega(L^{1/c^*})$: more embedding dimensions than reasoning steps.

$$\text{LLMs: } d_{\text{model}} \sim 10^3 \text{ to } 10^4 \ll L \sim 10^5 \text{ to } 10^6$$

Real models sit in the opposite regime. Variables cannot all be orthogonal, so interference adds to softmax leakage, the very quantity the theory controls. The authors flag this (footnote 4); Figure 1 stops at $L = 160$.

Experiment: one group, two actions, two code regimes

ACTIONS	S_5 acting on itself (free, NC ¹ -complete) · S_5 permuting $\mathbb{Z}_5$ (not free, NC ¹ -complete) · C_6 on $\mathbb{Z}_6$ (control)
CODES	One-hot variable codes (width $\geq X = 2,048$) versus learned codes of width $d_e \in \{32, 64, 128, 256\}$
PROTOCOL	One layer, no positional encoding, clause-level inputs; train at $L = 5$; test up to $L = 2,047$; log Acc _L and $m(L)$

$$m(L) = \text{Attn}_{\text{ans}, \ell \rightarrow \text{pred}, \ell+1} + \text{Attn}_{\text{ans}, \ell \rightarrow \text{ans}, \ell}$$

Predicted if the mechanism is right

S_5 on S_5 extrapolates like C_6 ; S_5 on $\mathbb{Z}_5$ stalls near twice its training length; $1 - m(L)$ grows linearly in L ; free-action horizons grow with d_e .

Challenges the result if

S_5 on S_5 stalls like S_5 on $\mathbb{Z}_5$: group complexity, not freeness, drives the gap. Or free-action horizons stop growing once $d_e \ll |X|$: the separation is an artifact of orthonormal codes.